

Autonomous Guided Vehicle Project

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Abstract

This report documents the design and development of an Autonomous Guided Vehicle (AGV) system in a simulated warehouse environment. It describes the current controller, earlier controller iterations, the logging infrastructure and the web application used to inspect simulation results. Replace this text with a concise summary of the final system, its main design choices and the outcomes of the evaluation.

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1 Current Controller

This section describes the current AGV controller implemented for the Webots warehouse simulation. The controller integrates perception, localization, planning and low-level motion control to move each vehicle through the environment while accounting for configured goals and dynamic obstacles. Unlike the monolithic scripts described in Section 2, the current implementation is organised as a set of cooperating modules behind explicit interfaces, so that each stage of the sense–think–act loop can be reasoned about, replaced or tested in isolation. Each vehicle in the warehouse runs an independent copy of this controller as a separate Webots process; there is no central coordinator, and all cooperation between vehicles is mediated indirectly through the shared world and the logging service described in section 3.

1.1 Architecture and Control Loop

The controller is driven by the `AGVSimulation` class in `DefaultController.py`, which owns one instance of each subsystem and executes a single fixed sequence once per simulation step. On construction the controller first establishes its identity: the Webots robot name is parsed by `parse_robot_unit_id` into a numeric `unit_id`, and a name that does not match the expected `PIONEER_3_<number>` pattern causes the controller to refuse to move rather than run with an ambiguous identity. This identifier is the key that ties a running vehicle to its configured route, its logging stream and its per-unit image outputs, and it is what allows several structurally identical controllers to coexist in the same world without interfering.

The controller depends on a small number of collaborators, each responsible for one concern. The `HardwareInterface` (implemented by `WebotsInterface`) wraps the Webots device API and exposes typed accessors for the motors, wheel encoders, GPS, inertial unit, lidar and camera, so that no control logic calls `robot.getDevice` directly. `Perception` converts raw device readings into a `SensorData` record; `Localization` turns that record into a filtered `RobotState`; `Task` supplies the current goal; the high- and low-level planners produce a route and a local path; `Control` converts the path into a motor command; and `RobotLog` records telemetry. The concrete planner classes are selected once, statically, through the `Task.hl_planner_class` and `Task.ll_planner_class` attributes, which currently bind to `VisGraphPlanner` and `GridPlanner` respectively. This indirection keeps the main loop agnostic to which planning strategy is in use and is what allows the alternative `SectorPlanner` (the direct descendant of the reactive avoidance logic discussed in Section 2) to be substituted without touching the control loop.

Each iteration of the loop, guarded by `hardware.is_alive()`, performs the same ordered pipeline. The controller reads the current simulation time, refreshes the task snapshot, perceives the environment, updates its state estimate, selects the active goal, plans a global route and then a local path, computes a motion command, applies the moving-obstacle safety limiter, checks the emergency-stop signal, and finally actuates the motors and captures telemetry. Figure 1 summarises this data flow. Ordering the stages this way enforces a strict information dependency: a command is only ever issued from a path that was planned against the most recent state estimate, which in turn was derived from sensor data captured in the same step.

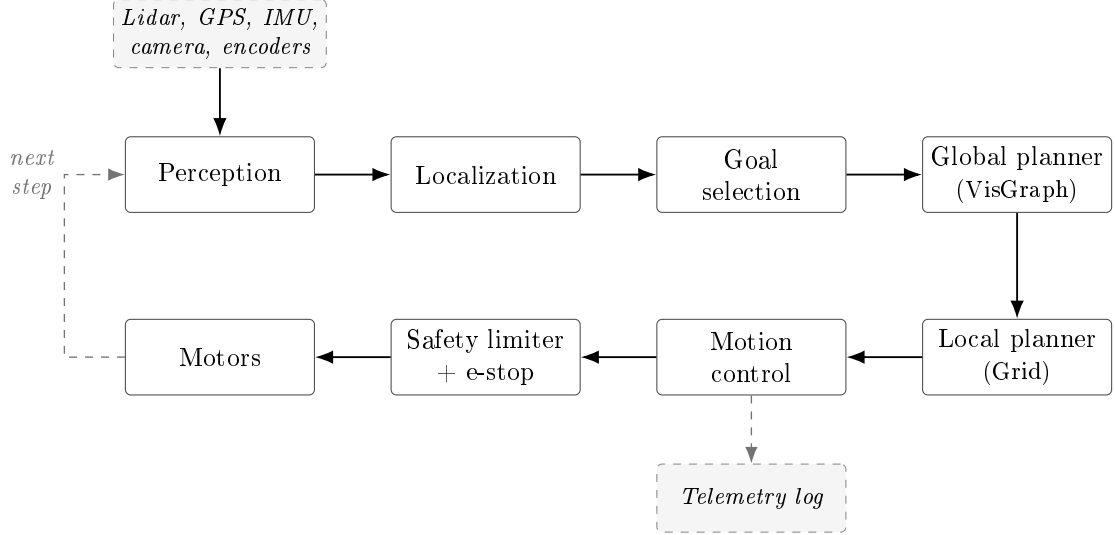


Figure 1: Per-step data flow of the current controller, read as a single pass from the sensors (top) back to the motors and round to the next step. Perception and localization produce the state estimate that both planners consume; the global route is refined into a local path, converted into a wheel command, and passed through the safety limiter and emergency-stop check before actuation, while telemetry is logged along the way.

1.2 Perception

The **Perception** module reads every sensor once per step and assembles the results into a single immutable **SensorData** snapshot, so that all downstream stages observe a consistent view of the same instant. The snapshot carries the wheel-encoder deltas, the raw GPS position, the inertial-unit yaw, the lidar point cloud and the camera image, together with the time delta since the previous perception call. Bundling these readings into one object avoids the class of bug where localization and planning inadvertently read sensors at slightly different times within the same step.

Object detection is an optional augmentation layered on top of the geometric sensors. When enabled through **VisionConfig**, an **ObjectDetector** runs a YOLOv8 “nano” network over the camera image and returns simple **Detection** records restricted to the person and forklift classes, with the network’s “truck” label remapped to “forklift” to match the warehouse domain. Because neural inference is far more expensive than the simulation timestep, detection is throttled to roughly ten frames per second and transparently uses a CUDA or Apple-Silicon accelerator when one is available, falling back to the CPU otherwise. The detector also writes an annotated camera frame to disk for the monitoring web application described in section 4. It is important to note the division of labour here: the lidar is the primary and always-present source of obstacle geometry, whereas vision only contributes *semantic labels* for obstacles the geometry pipeline has already found, and the controller remains able to navigate when detection is disabled or returns nothing.

1.3 Localization

The **Localization** module maintains the vehicle’s pose in the same absolute world frame used by the map and the GPS, producing a **RobotState** that records position, heading, linear velocity and angular velocity. Each step begins with a differential-drive odometry prediction: the left and right wheel-encoder deltas are converted into a centre displacement and a heading change, a small dead-band suppresses encoder noise, and the pose is advanced by integrating that motion from the previous estimate. This prediction alone would accumulate unbounded drift over a long

warehouse run, so it is corrected against two absolute references.

The correction is a weighted fusion with a disagreement safeguard rather than a plain average. For position, the module compares the predicted pose against the GPS reading: when the two disagree by more than `position_snap_threshold_m` the estimate is snapped directly to GPS, on the assumption that odometry has drifted too far to be trusted, and otherwise the two are blended using `gps_weight`. Heading is treated the same way, blending the inertial-unit yaw with the odometric heading under `yaw_weight` unless the angular disagreement exceeds `yaw_snap_threshold_rad`, in which case the estimate snaps to the measured yaw. This design keeps the estimate smooth during normal operation while still guaranteeing a bounded error, and it is a deliberate generalisation of the fixed, GPS-only fusion identified as a limitation of the earlier controllers in Section 2. The velocity fields are left as the encoder-derived prediction, since the absolute sensors provide position and heading but not a direct velocity measurement.

1.4 Goal and Route Management

Routes are data, not code. The `Task` module loads a validated snapshot from `goals.config.json`, which declares a set of named locations under `Goals` and, under `RobotRoutes`, an ordered list of goal names for each vehicle keyed by unit identifier. Loading is strict: the parser rejects malformed coordinates, duplicate goal names, non-numeric route keys and routes that reference an undefined location, and a vehicle whose unit has no configured route refuses to run. Externalising routes in this way decouples warehouse layout from controller logic, so pickup and dropoff points can be added or reordered without editing Python, and it is what makes the per-robot route editing exposed by the web application possible.

The task snapshot is re-read at the start of every control cycle through `Task.refresh`. If the file is temporarily unreadable or invalid at that moment, the controller logs a warning and retains the last valid snapshot rather than failing, which tolerates the brief inconsistency of an external tool rewriting the file mid-run. Goal advancement is handled in the loop by `__get_updated_goal`: when the distance from the state estimate to the active goal falls below `goal_reached_thresh`, the event is logged, the route index is advanced modulo the route length, and the next goal becomes active. Because the index wraps, each vehicle patrols its configured route indefinitely.

1.5 High-Level Planning

The high-level planner is responsible for the global question of *which way around the fixed warehouse structure* the vehicle should travel to reach its current goal, ignoring transient obstacles. It is implemented by `VisGraphPlanner` as a visibility-graph search over the static map. At construction the planner inflates every known obstacle rectangle by a fixed margin (`inflation`), takes the corners of the inflated rectangles as graph nodes, and precomputes which pairs of nodes are mutually visible, i.e. connectable by a straight segment that crosses no inflated obstacle. Inflating the obstacles before building the graph is what keeps the resulting routes clear of walls by roughly the vehicle’s footprint, so the local planner is rarely handed a global waypoint that sits inside structure.

At query time the planner first tests whether the goal is directly visible from the current position, in which case the route is the single goal point. Otherwise it connects the start and goal to their visible graph nodes and runs an A* search over the augmented visibility graph, returning the sequence of corners followed by the goal. Two robustness provisions matter in practice. First, if either the start or the goal has no visible graph node – for instance when the vehicle is momentarily inside an inflated region – the planner falls back to the nearest node reachable by an uninflated line of sight, so a route is still produced rather than an empty path. Second, the planner caches its last route and reuses it while the goal has not moved by more than `change_goal_thresh`, discarding only the waypoints the vehicle has already passed; this avoids rerunning the graph search every step and keeps the committed route stable instead of oscillating

between equal-cost alternatives. When no route can be found at all, the planner returns a degenerate path at the current position and logs a warning, leaving the vehicle stationary rather than moving blindly.

1.6 Local Planning and Obstacle Avoidance

The low-level planner turns the next global waypoint into a short, collision-free path through everything the vehicle can currently see, including obstacles absent from the static map. It is implemented by **GridPlanner**, which maintains a small egocentric occupancy grid centred on the vehicle and rebuilt at roughly ten hertz. The grid is world-aligned rather than body-aligned, which lets the planner convert between grid pixels and world coordinates without a rotation and simplifies reasoning about obstacle motion. Each replanning cycle proceeds in five stages: build the raw grid, separate static structure from unknown obstacles, track the unknown obstacles over time, inflate the grid with padding, and search for a path.

The raw grid is constructed by ray-tracing the lidar point cloud: cells along each beam up to its hit are marked free, the hit itself is marked occupied, and cells never touched by a beam remain unknown. Every occupied hit is then classified by *map differencing* against the global map: a hit that coincides with a known wall (within `tolerance_m`) is labelled static structure, while a hit with no counterpart in the map is treated as an *unknown obstacle*, the category that captures moving agents such as people and forklifts. This separation is central to the design, because static structure and dynamic obstacles are inflated differently in the next stage.

Unknown obstacle hits are passed through a cluster-track-label pipeline. The **ClusterExtractor** groups neighbouring hits into candidate objects, rejecting blobs too small to be real or too large to be anything but wall fragments. The **ObstacleTracker** associates clusters with persistent tracks across frames, maintains a short position history, and estimates each track’s velocity with exponential smoothing, promoting a track to “confirmed” only after it has been seen for several consecutive frames and classifying it as moving once its speed exceeds a threshold. Where a camera detection overlaps a track’s projected image position, the **YoloLabeler** attaches a sticky semantic label (person or forklift); tracks outside the camera frustum simply remain unlabelled and are still avoided on geometry alone. The padding stage then inflates the costmap: static structure and stationary obstacles receive a fixed circular margin, whereas a moving obstacle’s footprint is additionally stamped ahead of it along its estimated heading, so the vehicle yields space into which the obstacle is predicted to travel rather than only where it currently is.

Path search runs A* from the grid centre to the goal cell over this padded costmap, where free, unknown and padded cells carry increasing traversal costs so the planner prefers open, known space but will cross padding or unknown cells when necessary. The goal handed to the search is the global waypoint projected onto the boundary of the vision radius when it lies beyond the grid, and if the projected goal cell is not traversable the planner relocates it to the nearest admissible cell by sweeping outward around the vision circle. A subtlety preserved here is that the vehicle’s *final* route goal may sit inside conservative wall padding and still be accepted, but it is never accepted inside padding generated by a tracked obstacle or a peer vehicle, which prevents one AGV from parking on top of another. The resulting cell path is finally resampled at a fixed spacing by `__get_waypoints` into evenly spaced world-frame waypoints for the controller to follow.

Alongside the path, the planner publishes a **PlannerSafetyStatus** summarising the forward clearance to the nearest confirmed obstacle and to the nearest moving obstacle, measured from the vehicle footprint. This status is what couples perception to the motion-safety layer described next: it is computed only from confirmed tracks that lie ahead of the vehicle, so momentary or lateral detections do not needlessly halt progress.

1.7 Motion Control and Safety

The **Control** module follows the local path with a proportional controller. It takes the first waypoint as the immediate target, computes the distance error ρ and the heading error α between the state estimate and that waypoint, and derives a linear and an angular velocity from them. The linear velocity is scaled by $\cos \alpha$ so the vehicle slows when it is poorly aligned and drives at cruising speed when pointed at the target, the angular velocity is proportional to the heading error under **heading_p_gain**, and a small dead-zone suppresses angular jitter near zero error. Both velocities are clamped to configured maxima. A differential-drive conversion then turns the linear and angular velocity into individual wheel speeds and, if either wheel would exceed its limit, scales both down together so the requested curvature is preserved. This is the same actuation model validated in the earliest controller iterations and carried forward largely unchanged.

Two safety mechanisms sit between the raw command and the motors. The **MovingObstacleSafetyLimiter** consumes the planner's safety status and applies a two-tier rule: inside a slowdown clearance it caps the linear speed to a low crawl, and inside a tighter stop clearance it commands a full stop. The stop is *latched* – once triggered it is held for at least **minimum_stop_hold_s** and is only released when the forward clearance has recovered beyond a larger release threshold, which introduces hysteresis so the vehicle does not stutter between stopping and starting as an obstacle hovers near the boundary. Separately, a global emergency stop is honoured directly in the control loop: when an **emergency_stop.flag** file is present in the logging directory, the command is overwritten with zero wheel velocities. This file is written and removed by the monitoring web application (section 4) and provides an operator-facing kill switch that is independent of the planning and safety logic. Because the flag lives on disk, a stale file could otherwise persist across runs and hold a vehicle stopped indefinitely; the controller therefore deletes any existing flag once, at startup, so every run begins in a moving state and only an emergency stop raised during that run can halt it.

1.8 Observed Behaviour and Limitations

The layered design produces a clear division of responsibilities that is visible in operation: the visibility-graph planner commits the vehicle to a stable global route around fixed structure, while the grid planner continuously deforms the local path around people and forklifts that the static map does not contain, and the safety limiter provides a last line of defence when an obstacle enters the stopping clearance directly ahead. Separating semantic labelling (vision) from obstacle geometry (lidar) means the vehicle degrades gracefully: with object detection disabled it still avoids every obstacle, losing only the predictive, heading-aware padding applied to obstacles it can classify as moving.

Several limitations follow directly from the current design and are worth stating explicitly. Coordination between vehicles is entirely implicit: a peer AGV is perceived as just another unknown obstacle, with no communication of intent, so the fleet avoids collisions but does not actively deconflict routes, and a narrow corridor shared by two vehicles is resolved only reactively. The emergency stop, once raised during a run, is cleared only by the monitoring web application, so recovery from an operator halt depends on that external tool being available. Finally, the state estimate leans heavily on GPS being available and approximately correct; the snap-to-GPS safeguard bounds odometric drift but would itself propagate a large GPS error directly into the pose, since there is no third reference to arbitrate between the two.

The descriptions above characterise the controller's behaviour qualitatively, as it follows from the design and the code; this report does not attach numerical performance figures to them. A quantitative evaluation would measure the success rate over a full route, the frequency and cause of safety stops, the localization error against the Webots ground-truth pose, and any deadlock between vehicles sharing a corridor, and is the natural next step once representative simulation runs are recorded through the logging infrastructure of section 3.

2 Previous Controller Iterations

The current controller described in Section 1 was preceded by two implementations, `OldController_1` and `OldController_2`, developed in sequence. Each iteration validated a specific part of the sense-think-act loop before the project committed to the modular architecture used today. This section records their design choices, what they demonstrated, and the concrete limitations that motivated the redesign, so that the evolution of the system remains traceable.

Beyond the control logic itself, each iteration also shaped the simulated environment it was tested against, from a small static arena to a warehouse world that eventually included obstacles capable of moving independently of the robot. As discussed below, that last change was not cosmetic: introducing moving obstacles altered what the perception and obstacle-avoidance code in `OldController_2` actually had to guarantee, and is one of the clearest examples of the environment driving a design decision rather than the other way around.

2.1 Initial Approach

Development started on a Gctronic **e-puck** robot placed in a small test arena, and only later moved to a **Pioneer 3-DX** robot inside the full-scale warehouse world. This substitution was a change of simulated hardware and environment scale, independent of the two controller logic iterations described below: `OldController_1` was written for the e-puck arena, while `OldController_2` was written for the Pioneer 3-DX warehouse world that the project still uses today. The two robot models differ substantially in size, which is reflected directly in the kinematic constants each controller hardcodes: `OldController_1` uses a wheel base of 0.052 m and a wheel radius of 0.02 m, while `OldController_2` uses 0.33 m and 0.0975 m respectively, matching the Pioneer 3-DX manual. The change in scale is also visible in the goal coordinates each controller targets: `OldController_1` drives toward a single hardcoded point near (0.5, 0.5), consistent with a small test arena, whereas `OldController_2` cycles through a list of warehouse-scale waypoints such as (-29.3, 4) and (18.75, 2.25). Because the e-puck stage focused on validating the basic control and logging loop rather than warehouse-scale navigation, its lessons about obstacle-avoidance thresholds and acceleration behaviour do not translate directly to the Pioneer 3-DX scale, and had to be re-derived in `OldController_2`. The warehouse world itself grew in the same step, from a small empty arena to a layout with fixed shelving and, as described below, moving obstacles that `OldController_1` never had to contend with.

`OldController_1` implements a single-file, purely reactive controller whose purpose was to establish the minimal working pipeline: read the lidar and GPS, decide a wheel velocity, and persist a run history through the `RobotLog` service, tagged with `controller_version="old_controller"` so its runs remain distinguishable in the logs from later iterations. The only quantity it carries between simulation steps is the current wheel velocity read back from the motors; the controller keeps no explicit heading estimate. At every timestep it splits the 360-point lidar image, read by `read_lidar_image`, into three equal sectors (left, center, right) and computes the minimum range in each, and it derives a bearing toward the fixed goal directly from the raw GPS position with `get_angle`, rather than from any filtered state estimate. Sensor readings, wheel speeds and GPS coordinates are printed to the console roughly every half second, which was primarily a debugging aid for observing the controller live while the arena was small enough to watch directly. This console-first style of inspection was practical for a hand-sized arena and a single fixed goal, but it does not scale to a longer, multi-waypoint warehouse run in which relevant events are sparser in time and easy to miss in a scrolling terminal; this is part of the motivation for the more structured, periodically persisted logging that `OldController_2` adds, described later in this section.

Motion is selected through a fixed cascade of distance thresholds on the three sector minima: a full stop-and-reverse for a detected dead end, a hard turn for a close frontal obstacle, a gradual deceleration for a more distant frontal obstacle, and a small corrective turn when only a side sector

is triggered; in the absence of any obstacle the robot ramps up to a cruise speed. The `acc_speed` helper applies linear acceleration and deceleration ramps toward the selected target speed so that velocity commands do not change discontinuously between steps, and `robot_to_wheel_velocity` converts the resulting linear and angular velocity into individual wheel angular velocities using the wheel base and wheel radius, saturating both wheels together so that the requested curvature is preserved even when the maximum wheel speed would otherwise be exceeded. This approach proved effective for validating that lidar readings, GPS, motor commands and the logging pipeline (obstacle-state tracking, idle-time accounting, event logging) worked together correctly on a simple, single-goal task. It does not scale to the warehouse scenario, however: the goal is a single point hardcoded in the main loop, so the controller has no notion of a route or of multiple waypoints; obstacle avoidance is purely threshold-based on the current sector minima, with no memory of previously seen obstacles and no deliberate path planning, so the robot can only react once it enters an obstacle’s immediate vicinity rather than anticipate it; and the distance thresholds (for example 0.35 m for a frontal obstacle) were tuned for the e-puck’s scale and would need to be re-derived for a Pioneer 3-DX operating in a much larger environment, which is exactly what `OldController_2` did.

`OldController_2` restructures the control logic into a `RobotController_v1` class built around explicit `State` and `Position` dataclasses, and introduces a list of sequential warehouse waypoints (`goal_positions`) that the robot advances through and loops over once the final goal is reached. It reuses the same wheel-velocity conversion and saturation pattern as `OldController_1` in `set_robot_velocity`, adapted to the Pioneer 3-DX’s kinematic constants, and adds a camera device that is initialized and enabled but not yet consumed by the obstacle-avoidance logic, suggesting it was staged for a vision-based extension that was not completed in this iteration. Logging was extended accordingly: beyond the run history saved through `RobotLog`, the main loop periodically calls `save_to_database` roughly every five simulated seconds so that intermediate progress is persisted even if the simulation is interrupted.

The surrounding world was also extended with two `MovingWalls` instances, giving the controller dynamic obstacles to react to for the first time, in addition to the static warehouse layout. Each `MovingWalls` object locates a Webots node by its DEF name (`MOVING_WALL_1` and `MOVING_WALL_2`) through the controller robot’s Supervisor interface, records that node’s initial translation, and on every call to `move_wall` rewrites the corresponding coordinate of the node’s `translation` field as the initial position plus a sinusoidal offset with a configured amplitude and frequency along a chosen axis. In the main loop this method is invoked unconditionally at the very start of every simulation step, ahead of state estimation, planning and logging: the first wall oscillates along the y axis with an amplitude of 3 m, the second along the x axis with an amplitude of 5 m, both driven at the same angular frequency of 0.1 rad/s. Because `getFromDef` and direct field mutation are Supervisor-only operations in Webots, running this code requires the `OldController_2` process itself to hold Supervisor privileges, so a single script ends up simultaneously responsible for driving the vehicle and for choreographing part of the environment it drives through, a coupling that `OldController_1` never needed since its arena was entirely static.

Critically, `RobotController_v1` has no dedicated code path for the moving walls: from the lidar’s perspective a moving wall is indistinguishable from any other obstacle, and its motion only shows up indirectly, as occupied sector ranges that appear, disappear or shift between consecutive scans faster than a purely static obstacle’s would. The two `MovingWalls` are therefore the concrete scenario in this iteration that exercises the sector-clustering logic’s persistent-identity bookkeeping, described in the obstacle-avoidance discussion below, even though that logic itself does not treat static and dynamic obstacles any differently. In other words, the moving walls changed the requirements placed on the perception and avoidance code without requiring a single line written specifically for them: the obstacle-avoidance layer had to be correct for a world where sector boundaries can move on their own, and the walls were the mechanism by

which that requirement was actually tested during development.

State estimation in `OldController_2` combines wheel odometry, integrated through `update_wheel_odometry` and `update_with_odometry`, with GPS correction in `fuse_with_gps`. In practice this fusion is configured with a GPS weight of 1.0, so the corrected position is always the raw GPS reading and the odometry contribution is overwritten at every step rather than blended with it; the odometry-based prediction is computed but not actually used to stabilize the estimate. Heading and distance errors to the current goal are converted into linear and angular velocity commands by `calculate_velocity`, a proportional controller with gains `K_DISTANCE` and `K_HEADING` and a `cos(alpha)` term that slows the robot down when the heading error is large, with the resulting linear and angular velocities clamped to configured maxima and a small dead-zone applied to the angular command to avoid steady-state oscillation. Because this proportional scheme reacts only to the instantaneous heading and distance error, it carries no memory of previous corrections and no explicit model of the vehicle’s dynamics, which is consistent with the controller’s overall character as a direct, low-latency reaction to the current state estimate rather than a trajectory-tracking scheme.

Obstacle avoidance is the most developed part of this iteration. The lidar field of view is divided into 32 sectors, each classified as free or occupied, and consecutive occupied or free sectors are grouped into obstacle and free-space *clusters* that are assigned persistent identifiers across timesteps (`used_obstacle_ids`, `used_space_ids`, `previous_scan`), so that the same physical obstacle or gap keeps the same identity from one lidar scan to the next even as its exact sector boundaries shift. The two `MovingWalls` described above are the clearest source of this kind of shift: a purely static obstacle’s sector boundaries change only as the robot itself moves, whereas a moving wall’s boundaries also change on their own, independently of the vehicle’s trajectory, so the persistent-identity mechanism has to hold up under a case the static warehouse layout alone would not exercise. The controller then locks onto a specific free-space cluster (`locked_space`) while an obstacle blocks the direct path to the goal, which prevents it from oscillating between two competing free sectors on successive frames, and releases the lock once the obstacle is no longer relevant. When no free sector remains, the controller enters an escape mode that rotates in place until a free sector reappears. This sector-clustering strategy proved robust enough that, as discussed below, it was carried forward almost unchanged into the current controller.

The implementation also exposes concrete limitations. The persistent-identity logic that tracks clusters across frames is intricate and stateful, and the code raises an exception if a sector is left without an assigned identifier, which is a fragile failure mode for a long-running simulation. The file still contains earlier, commented-out obstacle-avoidance attempts (`old_obstacle_avoidance`, `min_distances`), which indicates that the sector-based strategy was reached iteratively rather than designed upfront. Goal waypoints and robot-specific parameters remain hardcoded in the controller script instead of being externalized to a configuration file. The Supervisor privileges required to animate the `MovingWalls` are a further limitation: granting a per-robot controller direct write access to arbitrary scene nodes is difficult to justify once several AGVs share the same warehouse, since any one controller process could in principle reposition world geometry that every other vehicle relies on for localization and obstacle avoidance, and `OldController_2` does not scope or guard that access in any way.

2.2 Lessons Learned

Comparing the two iterations with the current controller (Section 1) shows a clear line of continuity rather than a full rewrite. The sector-clustering obstacle-avoidance algorithm introduced in `OldController_2` is, in its essential structure, the same algorithm implemented today by the `SectorPlanner` low-level planner: the same notion of dividing the lidar field of view into sectors, assigning persistent obstacle and free-space identifiers across frames, locking onto a chosen free sector, and rotating to escape a dead end. This is direct evidence that the local, reactive avoidance layer designed and tuned in `OldController_2` was robust enough to be promoted into

production, once decomposed behind a dedicated planner interface and driven by configuration objects instead of hardcoded constants. The `MovingWalls` introduced in `OldController_2` were, in this sense, an early stress test for that design: because they could enter or leave a sector independently of the robot’s own trajectory, they exercised the persistent-identity bookkeeping in a way a purely static layout would not, and the fact that the same clustering approach was carried forward into the `SectorPlanner` suggests it held up under that test rather than being reworked to cope with dynamic obstacles later. The lower-level wheel-velocity conversion pattern first written in `OldController_1` and reused in `OldController_2` survived the transition as well, which suggests that the basic differential-drive actuation model was correct from the earliest iteration and needed no substantial rework, only re-parameterization for each robot’s physical dimensions.

Other aspects of the design changed more substantially. Route information moved out of the controller code entirely: instead of the hardcoded `goal_positions` list, waypoints and per-robot routes are now declared in `goals.config.json` as named locations, which decouples warehouse layout changes from controller logic and lets new pickup or dropoff points be added without touching Python code. The monolithic `RobotController_v1` class was split into dedicated perception, localization, planning and hardware modules, and a high-level planner was introduced above the reactive sector layer so the vehicle now follows a globally planned route rather than only locally reactive corrections. Hardware access was also wrapped behind a dedicated hardware-interface layer instead of calling `robot.getDevice` directly from control logic, improving testability and making the sensor and motor interfaces explicit; the camera device that `OldController_2` enabled but never used is, in this later architecture, finally exercised by a dedicated perception module.

The GPS/odometry fusion limitation identified in `OldController_2` was also addressed directly rather than carried forward silently: the current localization module generalizes the fixed `gps_weight = 1.0` into a configurable weighted fusion, adds a disagreement threshold that snaps the state estimate to GPS only when odometry has drifted too far, and additionally fuses a heading estimate from the inertial unit, which the old controllers did not use at all. Taken together, these observations show that the old controllers were most useful not as discarded prototypes but as a way to isolate which parts of the design were already sound and which parts still needed rework: the sector-based reactive avoidance layer and the differential-drive actuation model proved robust enough to keep, while state estimation, goal configuration and, most importantly, coordination between multiple AGVs sharing the same warehouse still required further work before the system could operate reliably at full scale.

3 Logging Infrastructure

The logging subsystem provides a persistent record of simulations, events and telemetry generated by the AGVs. Its role is to make the execution of a simulation inspectable after the fact and to provide the data consumed by monitoring tools. The pipeline is built around a client-server architecture: each robot controller collects events in memory and periodically flushes them to a standalone HTTP service backed by MySQL.

3.1 Service Architecture

The logging pipeline follows a client-server model. Each Webots robot controller process runs an instance of the `RobotLog` class in `logging/logger/robot_log.py`, which acts as an in-memory event collector and HTTP client. A separate process hosts the logging server defined in `logging/logging_server.py`, which exposes a set of HTTP endpoints split between write and read responsibilities, all served by Python’s `ThreadingHTTPServer`. Using a thread-per-request model allows the server to handle concurrent flush operations from the two robots deployed in

the warehouse world and read requests from the monitoring web application without blocking any client.

The write-side and liveness endpoints serve distinct roles in the simulation lifecycle:

- **POST /start** accepts a `unit_id` string, creates a new row in the `Simulations` table, and returns the assigned `sim_id` to the client. This identifier anchors every subsequent event and telemetry record for the run.
- **POST /save** accepts a batch payload containing updated summary fields, a list of events and a list of telemetry records. The server validates the payload, opens a database transaction, and atomically writes all three components to MySQL. An HTTP 200 response indicates success; a 400 or 500 response signals validation or infrastructure failure, respectively.
- **GET /health** returns `{"status": "success"}` and serves as a liveness probe for the caller.

The read side of the API serves the monitoring web application described in section 4. Five **GET** endpoints expose committed simulation data: `/simulation-options` lists the available unit identifiers together with their simulation identifiers; `/simulations`, `/events` and `/event-telemetry` return rows from the corresponding database tables; and `/database-snapshot` returns all three collections in a single consistent response. Each filtering endpoint accepts optional `unit_id` and `sim_id` query parameters that restrict the result to one robot or one run. The design, validation and concurrency handling of these endpoints are detailed in section 3.6.

The server listens on `127.0.0.1:8080` by default. Command-line arguments `-host` and `-port` override the listener address, while the `LOGGING_SERVER_URL` environment variable controls the URL used by the robot controllers. This layered configuration supports local development, where the server and both controllers run on the same machine, and any future extension to a distributed setup.

3.2 Data Model

The server persists simulation data in three MySQL tables created by the queries in `Database_Structure.sql`. Each table addresses a distinct level of granularity: run-level aggregates, discrete event records, and per-event controller state snapshots. Figure 2 summarises the three tables, their columns and the relationships between them; the remainder of this subsection describes each table in turn.

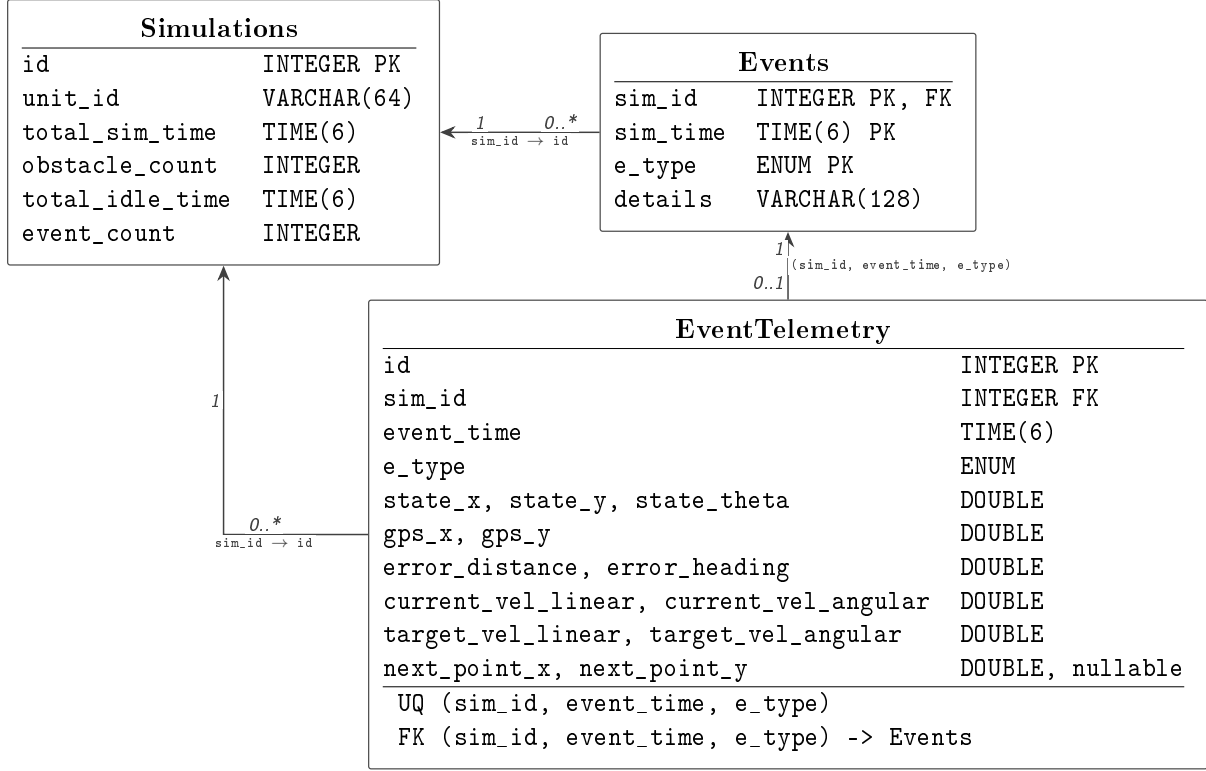


Figure 2: UML-style schema of the logging database. Each arrow points from a foreign key in the child table to the key that it references in the parent table. Multiplicities describe the number of child rows associated with one parent row. PK, FK and UQ denote primary-key, foreign-key and unique constraints, respectively.

Simulations. The **Simulations** table holds one row per completed run. Its auto-increment **id** serves as the primary key and is the foreign key target of the event-related tables. Each row stores the robot’s **unit_id** as a variable-length string and four aggregate metrics: **total_sim_time**, **obstacle_count**, **total_idle_time** and **event_count**. Temporal columns use the MySQL **TIME(6)** type with microsecond precision, preserving sub-second resolution for runs that span tens of minutes.

Events. The **Events** table records discrete state transitions and notable occurrences. Its composite primary key (**sim_id**, **sim_time**, **e_type**) ensures that a simulation cannot contain two events of the same type at the same instant. The **e_type** column is modeled as a MySQL **ENUM** restricted to eight values:

- **START** — logged when the controller confirms a new simulation record.
- **STOP** — logged during controller shutdown.
- **IDLE_START** and **IDLE_END** — mark transitions when the robot’s linear velocity crosses a near-zero threshold.
- **OBSTACLE_ENCOUNTER** and **OBSTACLE_CLEARED** — reflect the presence or absence of dynamic obstacles detected by the low-level safety module.
- **REACHED_TARGET** — recorded when the controller’s distance to the current goal falls below the configured threshold.
- **UNEXPECTED_BEHAVIOR** — an open-ended category for anomalous conditions observed by the controller.

Each event carries a `details` column of up to 128 characters. For obstacle events the details encode the GPS coordinates at which the obstacle was detected or cleared; for idle transitions they record the linear and angular velocity that triggered the state change; for target events they include the goal index and coordinates. This compact textual context makes the event log self-describing when inspected without joining against telemetry.

EventTelemetry. The `EventTelemetry` table extends each event with a complete snapshot of the controller’s internal state. Its 14 numeric columns capture the estimated pose (`state_x`, `state_y`, `state_theta`), the GPS reading (`gps_x`, `gps_y`), the control errors computed from the current pose to the goal (`error_distance`, `error_heading`), the current linear and angular velocity (`current_vel_linear`, `current_vel_angular`), the command velocities output by the control module (`target_vel_linear`, `target_vel_angular`), and the coordinates of the next waypoint on the low-level path (`next_point_x`, `next_point_y`). The waypoint fields are nullable because the planner may produce an empty path under certain conditions; all other fields are non-nullable `DOUBLE` columns.

A `UNIQUE` constraint on (`sim_id`, `event_time`, `e_type`) mirrors the `Events` composite primary key, allowing each event to have at most one telemetry row. The relationship is therefore one-to-zero-or-one from `Events` to `EventTelemetry`: a telemetry row must reference an event, while the schema does not require every event to have telemetry. A foreign key from (`sim_id`, `event_time`, `e_type`) to the corresponding columns in `Events` enforces this referential integrity.

3.3 Data Flow and Controller Integration

The logging lifecycle is tightly coupled to the controller’s execution in `AGVSimulation.run()` (`DefaultController.py:74`). During initialisation, the constructor instantiates a `RobotLog` with the configured server URL and the robot’s `unit_id`, then calls `logger.start()`. This method sends a `POST /start` request and, upon success, stores the returned `sim_id` and logs a `START` event on the client side. If the server does not respond successfully within ten attempts, spaced one second apart, the method returns `False`. The controller responds by calling `_refuse_to_run()`, which stops the motors and leaves the robot immobile for the remainder of the simulation. This design ensures that no robot operates without a persistent audit trail.

Once the simulation is running, the controller’s main loop executes at the Webots step rate. On every cycle, three logging operations take place:

1. `capture_telemetry()` (`robot_log.py:233`) stores the latest state estimate, GPS reading, control command and low-level waypoint in the `latest_telemetry` dictionary. This snapshot is held in memory and attached to the next event that is logged. The method also receives the current simulation time and forwards the same data to the realtime mirror described in section 3.5, which writes a live snapshot of the robot’s state to disk for the monitoring dashboard.
2. `update_obstacle_state()` (`robot_log.py:297`) compares the current obstacle presence flag from the low-level planner’s safety status with the previous state. A transition from clear to obstructed produces an `OBSTACLE_ENCOUNTER` event; the reverse transition produces `OBSTACLE_CLEARED`. The obstacle count is incremented once per encounter, not per cycle, giving a meaningful count of distinct obstruction episodes.
3. `update_idle_state()` (`robot_log.py:311`) detects transitions based on whether the robot’s linear velocity magnitude falls below the threshold `idle_speed_threshold` (default 10^{-3}). It accumulates the total idle time and logs `IDLE_START` and `IDLE_END` events at each transition.

Additionally, `log_target_reached()` is called from the controller’s goal-handling logic (`DefaultController` when the distance error to the current target drops below `goal_reached_thresh`. This produces

a `REACHED_TARGET` event that includes the goal index and coordinates.

Rather than sending each event individually, the `RobotLog` accumulates events and their matching telemetry records in Python lists and flushes them in periodic batches. The flush interval is controlled by `LogConfig.log_interval` in `config.py`, set to 5 seconds by default. The decision to batch events rather than stream them avoids saturating the HTTP server with a request per simulation step (which runs at hundreds of hertz) and reduces the number of database transactions.

The batch flush procedure in `RobotLog.flush()` (`robot_log.py:356`) uses a two-queue design. The `failed_events` and `failed_event_telemetry` lists hold records that could not be persisted in a previous attempt. On each flush, the failed queue is retried first by calling `_send_batch()`. If the retry succeeds, the failed queues are cleared and the current buffer is sent. If the retry fails, the current buffer is appended to the failed queues, preserving the chronological order of events across multiple failed attempts. Only after a successful round-trip are events removed from both the main and failed queues.

When the simulation terminates, the `finally` block of `run()` (`DefaultController.py:143`) calls `logger.stop()`, which logs a `STOP` event and enters a retry loop for flushing all remaining data, including the failed queue. If the server remains unreachable after ten attempts, the method prints the number of unsaved records and returns `False`, accepting controlled data loss at a well-defined shutdown boundary. A nested `finally` block then calls `logger.clear_realtime()` (`robot_log.py:160`) to truncate the unit's live snapshot file, so the monitoring dashboard does not continue to display a stale pose for a robot that is no longer running.

3.4 Validation and Transactional Integrity

The server enforces payload correctness before any data reaches the database. The validation function `validate_save_payload()` (`logging_server.py:120`) performs a comprehensive check of every field in the request body. Numeric fields must be finite; integer fields must be of type `int` and nonnegative; string fields must respect length limits; the `e_type` of every event and telemetry record must belong to the `ALLOWED_EVENT_TYPES` frozenset; and the `sim_id` embedded in each sub-object must match the top-level `sim_id`. The validator also constructs sets of event identities, defined as tuples of `(sim_id, time, e_type)`, and verifies that the identities from the `events` and `event_telemetry` arrays match exactly and contain no duplicates. This cross-consistency check catches misalignments between the two arrays before the database transaction begins.

When validation fails, the function raises `PayloadValidationError` with a message identifying the offending field. The HTTP handler catches this exception and returns a 400 response, allowing the controller to distinguish between malformed data (which cannot be fixed by retrying) and infrastructure failures (which may be transient).

The `PersistenceService.save()` method (`logging_server.py:465`) writes accepted data inside a single database transaction. It first acquires a row-level lock on the target `Simulations` row with `SELECT ... FOR UPDATE` and verifies that the stored `unit_id` matches the requesting unit. This ownership check prevents a controller from accidentally or maliciously overwriting another robot's simulation data. The method then updates the summary columns and inserts events and telemetry rows using `executemany()` with `ON DUPLICATE KEY UPDATE` clauses. The idempotent insert makes the operation safe to retry: if a previous flush succeeded but the response was lost, re-sending the same batch will not create duplicate rows.

If any database operation raises an exception during the transaction, the entire batch is rolled back and a `PersistenceError` is raised, which the HTTP handler maps to status 500. The `save()` method treats `PayloadValidationError` specially: it rolls back the transaction but does not wrap the exception, allowing the handler to return status 400. This separation of error classes keeps the controller informed about the nature of each failure.

Concurrency is managed at two levels. The `PersistenceService` uses a `threading.Lock` to

serialise every `save()` and read method, ensuring that transactions from concurrent flush operations by the two robots and concurrent read requests do not interleave. The MySQL connection is not held open between flushes; instead, `_get_connection()` (`logging_server.py:226`) checks the liveness of any cached connection with `ping(reconnect=True)` and creates a new connection when needed. This lazy-reconnect strategy avoids the complexity of connection pooling for a workload where writes arrive no more than every five seconds and reads are short `SELECT` statements.

3.5 Realtime Telemetry Mirror

The batched flush design described in section 3.3 makes committed data available to the database only once every few seconds, and always at least one flush interval behind the controller’s current state. For the monitoring web application to render a live view of the robots in motion, it needs a more recent signal than the committed record can provide. The `RobotLog` class therefore maintains a second, independent output channel: a single-record JSONL file that always reflects the robot’s latest controller cycle.

Each time `capture_telemetry()` is called, it delegates to `_write_realtime_snapshot()` (`robot_log.py:117`), which serialises the current simulation time, estimated pose, GPS reading, control errors, current and target velocities, goal position and next waypoint into a compact JSON object. The payload is written to a file named `robot_controller_runs_{unit_id}_realtime.jsonl` in the `logging/logs` directory, where `unit_id` is the robot’s numeric identifier. Rather than appending, the writer replaces the file contents atomically: it writes to a temporary file in the same directory and then calls `os.replace()` to swap it into place. This is the same temporary-file-and-rename pattern used by the server-side JSONL mirror (section 3.7), and it guarantees that a reader never observes a partially written record.

Writing on every controller cycle would exceed the refresh rate the dashboard can usefully display and would add avoidable disk traffic. The writer therefore throttles updates with `REALTIME_WRITE_INTERVAL` (`robot_log.py:11`), set to 0.05 s by default, so that the snapshot is refreshed at most twenty times per second. The throttle compares successive simulation timestamps rather than wall-clock time, keeping the update rate tied to simulation progress. A non-numeric `unit_id` disables the mirror entirely, since the filename scheme assumes a numeric identifier; the writer prints a diagnostic and returns `False` in that case.

The file is intended to hold exactly one line, representing the most recent state. When the controller shuts down, the nested `finally` block in `AGVSimulation.run()` calls `clear_realtime()` (`robot_log.py:160`), which overwrites the file with empty content using the same atomic-replace primitive, so that the dashboard does not keep displaying a robot that has stopped. Because the realtime mirror is written by the controller process and read from disk, it is independent of both the HTTP server and the database: a database outage does not interrupt the live view, and the live view does not compete for database locks. The trade-off is that the mirror carries only the current state, not a history. Once a snapshot is overwritten the previous state is lost, and the committed event and telemetry tables remain the authoritative record of past behaviour.

3.6 Server-Side Read API

The logging server exposes five `GET` endpoints that let clients retrieve committed data without direct access to MySQL. They serve the monitoring web application’s historical and tabular views, complementing the live realtime mirror described in section 3.5. All read endpoints share the same `ThreadingHTTPServer` and `PersistenceService` instance as the write path, and they are dispatched by the `do_GET` handler (`logging_server.py:645`).

Endpoints and filtering. `GET /simulation-options` takes no parameters and returns the set of `unit_id` values present in the `Simulations` table, each paired with the list of its simulation

identifiers ordered most-recent-last. It lets the dashboard populate unit and run selectors before issuing more specific queries. `GET /simulations`, `GET /events` and `GET /event-telemetry` return rows from the corresponding tables, while `GET /database-snapshot` returns all three collections in a single response; the last is the endpoint the web application uses for its tabular view, since one request yields a self-consistent picture of a run. Every filtering endpoint accepts optional `unit_id` and `sim_id` query parameters, validated by `validate_read_query()` (`logging_server.py:76`), which rejects any other parameter name, requires each filter to appear exactly once, and normalises both values to numeric strings or positive integers. `validate_empty_query()` (`logging_server.py:104`) enforces that `/simulation-options` and `/health` receive no query string. As on the write path, a `PayloadValidationError` maps to HTTP 400 and a `PersistenceError` to 500.

Result limiting and ordering. Each read query is capped at `DEFAULT_READ_LIMIT` (`logging_server.py:17`) set to 200 rows. The underlying `SELECT` statements order by the primary identifier in descending order and apply a `LIMIT` clause, after which the result list is reversed in Python. This combination returns the most recent 200 rows in chronological order, bounding the response size for endpoints that could otherwise return the full history of every run. The same limit applies to the individual endpoints and to `/database-snapshot`.

Consistency and concurrency. The `/database-snapshot` endpoint is served by `PersistenceService.read()` (`logging_server.py:413`), which opens a single transaction with `start_transaction(consistent_snapshot=True, readonly=True)` before issuing the three `SELECT` queries. Because all three reads execute inside one consistent read snapshot, the simulations, events and telemetry arrays cannot diverge even if a concurrent flush commits new data between the individual queries. The other read endpoints issue a single query each and therefore need no snapshot transaction. Like the write path, every read method acquires the `PersistenceService` instance lock and reuses the cached MySQL connection through `_get_connection()`. Reads never commit: the `finally` block of each method calls `_rollback()` to end the read transaction and release the snapshot, leaving the connection ready for the next request.

Response caching and serialisation. Read responses carry a `Cache-Control: no-store` header, set by the `no_store` flag in `_send_json()` (`logging_server.py:722`), which instructs the browser and any intermediate proxy not to cache the payload. This prevents the dashboard from displaying stale data when a user refreshes a view between simulation updates. The same method serialises responses with `PersistenceService._json_default()` (`logging_server.py:632`), which converts the `timedelta`, `datetime` and `time` objects returned by the MySQL connector into strings and `Decimal` values into floats, so that temporal columns such as `total_sim_time` and `sim_time` survive the JSON round-trip without manual conversion in the caller.

3.7 JSONL Mirror and Dashboard Integration

After every successful database commit, `save()` invokes `_best_effort_rebuild()` (`logging_server.py:560`), which in turn calls `rebuild_jsonl()` (`logging_server.py:566`) to regenerate three flat files in the `logging/logs` directory: `simulations.jsonl`, `events.jsonl` and `event-telemetry.jsonl`. Each file contains one JSON object per line, with keys derived from the MySQL column names. The rebuild issues its `SELECT` queries on the cached connection after commit, reading from a consistent snapshot of the committed state, so the files faithfully represent the data just written. The call is wrapped in a best-effort guard: if file regeneration raises, the server logs a warning but leaves the already-committed database rows intact.

The files are written atomically using a temporary-file-and-rename pattern (`_write_jsonl_atomic()`, `logging_server.py:610`). Data is written to a temporary file in the same directory, flushed to

disk with `os.fsync()`, and then moved to the final destination with `os.replace()`, which is atomic on the same filesystem. If any exception interrupts the process, the temporary file is removed. A reader therefore always sees a complete, self-consistent snapshot rather than partial content.

This mirror serves primarily as a durable, human-readable archive of every simulation run. A developer can inspect the latest simulation with `tail simulations.jsonl`, search for specific event types across all runs with `grep`, or import the data into a Python notebook for statistical analysis, without navigating the MySQL interface. The monitoring web application does not read these files directly: its live view consumes the client-side realtime mirror of section 3.5, while its committed-data views go through the read API of section 3.6. The archive and the read API nevertheless present the same committed state, since both are derived from the database; the archive optimises for ad-hoc human inspection, the API for structured client consumption.

Together the two read channels cover the dashboard’s two temporal modes. While a simulation is in progress, the realtime mirror offers a low-latency view of each robot’s current pose and velocity, read from disk without involving the logging server or the database. Once data has been flushed and committed, the read API provides the consistent, filtered snapshots used by the historical and tabular views. This separation keeps the live display responsive and the committed display accurate, at the cost of a brief window in which the live view may be ahead of the last committed flush.

3.8 Reliability Considerations

The logging system adopts a conservative approach to reliability that prioritises data completeness over availability during startup while tolerating transient failures during steady-state operation.

The most consequential decision is the controller’s refusal to move if the logging service cannot create a simulation record at startup, enforced in `RobotLog.start()` (`robot_log.py:211`). Ten consecutive HTTP failures are treated as a terminal condition; the robot remains immobile for the entire simulation. This stop-the-world posture is chosen because a missing simulation record would render all subsequent events unanchored. The `Events` and `EventTelemetry` tables carry foreign keys referencing `Simulations.id`, and without a valid `sim_id` no event data can be committed to the database. Buffering events indefinitely in hope of a late recovery would risk unbounded memory growth and opaque data loss, making post-hoc debugging harder.

During normal operation the batch-flush design trades some timeliness for robustness. Events spend up to five seconds in controller memory before being sent to the server. A crash of the controller process or Webots within a flush interval loses at most the events accumulated since the last successful flush. The failed-events queue absorbs transient server unavailability: if the server is down when a batch is attempted, the records are preserved and retried on the next interval without interrupting the simulation loop. This decoupling ensures that a brief database outage does not halt the robots, at the cost of potentially losing the in-memory batch if the controller itself terminates before the next successful flush.

At shutdown, the `RobotLog.stop()` method makes up to ten additional flush attempts. Data that remains unsaved after these attempts is discarded, and a console message reports the count. This explicit best-effort-accept-loss boundary avoids blocking the application exit indefinitely while making the extent of any data loss visible to the operator.

The MySQL container is provisioned via Docker, as documented in the project `README.md`. The database connection parameters (host, port, user, password, database name) are read from environment variables with defaults that match the Docker setup, allowing the server to be reconfigured without code changes. The container runs on the standard MySQL port 3306 and stores its data in a Docker volume that persists across container restarts.

3.9 Use in Evaluation

The structured event log is the primary source of quantitative evidence for evaluating controller behaviour. The **EventTelemetry** table records the robot’s full estimated state at every logged moment, enabling offline reconstruction of trajectories, analysis of convergence toward goals, and comparison of planning decisions across different controller configurations or environment layouts.

The summary metrics aggregated per simulation run (**total_sim_time**, **total_idle_time**, **obstacle_count**, **event_count**) support direct quantitative comparison between runs. For example, total idle time can serve as a proxy for navigation efficiency: a robot that waits for obstacles to clear contributes more idle time than one that finds alternative routes. The obstacle count reflects the frequency of dynamic interference, which varies with the movement patterns of the supervisor-controlled humans and forklifts in the environment. Because each simulation row is identified by its **unit_id**, results from the two robots operating concurrently can be compared side by side, controlling for the same world conditions.

The **details** column of the **Events** table provides a lightweight narrative of each run. Searching for **OBSTACLE_ENCOUNTER** events reveals the locations and frequencies of obstruction episodes; inspecting the GPS coordinates in the details string can identify bottlenecks in the warehouse layout where dynamic obstacles cluster. **UNEXPECTED_BEHAVIOR** events capture anomalous conditions that the controller explicitly flags, making them immediately searchable without scanning the full telemetry.

The JSONL mirror enables ad-hoc analysis with standard command-line tools. A developer can extract all events of a given type with **grep**, compute aggregate statistics from the telemetry file with a short script, or load the data into a Jupyter notebook for more involved analysis. The web application described in section 4 renders the monitoring dashboard from the read API and the realtime mirror described above, while the JSONL archive remains available for direct programmatic inspection. Both interfaces are derived from the same committed database state, so ad-hoc analysis and visual oversight stay consistent with one another.

4 Web Application

The current Web App is a React/Vite prototype designed for local operational use during AGV simulations. It provides a unified interface for observing active robots, managing their configured routes, and inspecting data retained from simulation runs. This subsection describes the application from the operator’s perspective; its internal structure and its role within the wider system are addressed separately in the subsequent subsections.

4.1 User Interaction with the Web App

4.1.1 Startup, Prerequisites, and Access

Initial preparation requires the project Python environment to be created with **./setup.sh**, the frontend dependencies to be installed with **npm install** from **web-app/**, and Webots to be configured to use the Python interpreter created by the setup script. The MySQL instance must also contain the schema defined in **Database_Structure.sql**. These steps establish the runtime dependencies needed by the controllers, the logging service, and the Web App.

For normal operation, MySQL is started first, followed by **logging/logging_server.py** and then the Webots warehouse simulation. This order is operationally significant because each **DefaultController** must create a simulation record through the logging service before it is permitted to move. The Web App is started from **web-app/** with **npm run dev**; it may also remain open between simulation runs because active robot units are discovered automatically as their data becomes available.

The interface is accessed through the local address printed by Vite, normally `http://localhost:5173`. In the current implementation, the application APIs are registered by the Vite development server, so `npm run preview` does not provide the complete operational interface. The system should therefore be regarded as a local operational application and a prototype rather than as a production deployment.

4.1.2 Interaction Model and Available Features

On desktop displays, the principal pages are reached through the sidebar and top navigation, whereas compact displays use a bottom navigation bar. A fixed page-control dock presents only the selectors and refresh settings relevant to the current view. The Robot selector chooses the unit shown by single-robot pages, while the Simulation selector restricts database-backed information to a specific run or retains the broader *All simulations* view.

The available views and their principal interaction mechanisms are summarized in table 1. The table distinguishes live operational information from data retrieved from MySQL, since the two sources serve different user tasks and are refreshed independently.

Table 1: Operational views of the current Web App.

View	Primary purpose	Principal information	Page controls
Dashboard	Monitor and operate one selected robot	Mission status, telemetry, configured route, warehouse map, robot-produced JPEG images, and compact MySQL records	Robot, Simulation, Live refresh, Database refresh
Map View	Examine the selected robot spatially	Enlarged warehouse map with optional grid, obstacle, path, and goal layers; camera and local-planner images	Robot, Live refresh
Fleet Status	Compare all currently discovered units	One live status card per unit, including motion, navigation, and error information	Live refresh
Log Analysis	Review retained simulation data	Simulation, event, and event-telemetry tables; event filtering; linear-velocity and heading-error charts	Robot, Simulation, Live refresh, Database refresh
Help Center	Provide embedded operating guidance	The rendered content of <code>docs/web-app-user-guide.md</code>	None

Live refresh can be configured between 50 ms and 1 s, with 50 ms as the default, and controls requests for unit-specific real-time JSONL data and robot-produced JPEG images. Database refresh can be configured between 3 s and 30 s, with 10 s as the default, and controls the retrieval of MySQL snapshots. The selected robot and both refresh intervals are retained by the browser between visits, whereas the selected simulation is local to the current session.

The Dashboard is the principal single-robot workspace. It combines the mission runtime, estimated navigation state, current goal, next local-planner point, current and target velocities, position and control errors, route configuration, warehouse visualization, and the two robot-produced image streams. It also includes compact views of the selected robot’s simulations,

events, and event telemetry, thereby allowing live behavior and retained records to be inspected without changing page.

The Map View provides a larger spatial representation and permits the user to toggle the grid, obstacles, planned path, and goals, as well as to change the zoom level. Fleet Status instead removes the single-robot selector and creates a summary card for every discovered unit, supporting simultaneous observation of independently operating robots. Log Analysis concentrates on database-backed inspection through tables, event-type filtering, and charts derived from at most the latest 100 event-telemetry rows for the selected simulation; when all simulations are selected, the charts use the highest available simulation identifier for the chosen robot.

4.1.3 Main Operational Workflow

After the required services and the Webots world have been started, the operator opens the local Web App and waits for active units to be discovered. A robot is then selected for the Dashboard or Map View, and a simulation may additionally be selected on database-backed pages. The operator can follow the live mission through the map, telemetry, planner state, and image outputs, while the independent database refresh provides progressively updated records of the run.

Goal management is available from the Dashboard and is intended for use during a live simulation. For the selected robot, the operator can move goals earlier or later in the route, append a uniquely named goal with finite coordinates, or remove a goal. The interface prevents deletion of the final route goal and preserves goals shared with another route; accepted changes update `AGV_Webots_World_and_Controllers/controllers/DefaultController/goals.config.json`, which the controller reloads while operating. Concurrent manual editing of this file should be avoided while changes are being saved through the Web App.

The desktop sidebar also exposes a global control labelled *Emergency Stop*. In architectural terms, this control is an application-level global motion inhibit: it applies to every active robot using `DefaultController`, causes motion commands to be replaced by zero linear, angular, and wheel velocities on the next controller cycle, and leaves the Webots simulation running. The inhibited state remains active until the operator selects *Resume Robots*; it is global rather than associated with the robot currently selected in the interface.

Once immediate monitoring is complete, the operator can use Log Analysis to review the selected unit and simulation in greater detail. This progression from unit discovery, through live observation and optional route management, to database-backed review constitutes the principal operational workflow supported by the current prototype.

4.2 Web App Architecture

4.2.1 Current State and Architectural Style

The active Web App is a single-page React application written in TypeScript and built with Vite. Its declared runtime dependencies in `web-app/package.json` are React, React DOM, React Router, Lucide React, and locally packaged Inter and JetBrains Mono fonts; Vite, the React plugin, and TypeScript provide the development and build toolchain. The application is operational for local simulation monitoring, but its server-side capabilities are installed through Vite's `configureServer` hook and are therefore available only when the development server is running. Consequently, the implemented functions constitute a local operational prototype, while independently deployable production serving is not implemented.

The implementation follows a compact layered arrangement rather than a conventional separation into independently deployed frontend and backend services. The browser layer renders the interface and holds transient application state, client modules validate and request data, and custom Vite middleware provides a same-origin API over project files and the logging service. This organization reduces the number of runtime processes required for local operation and prevents

the browser from accessing the filesystem directly. Its main trade-off is that Vite simultaneously acts as build tool, static application server, and lightweight backend-for-frontend.

The previous Streamlit implementation remains in `web-app/app.py`, but it is a predecessor and is not part of the current application architecture.

4.2.2 Frontend Composition

The HTML document in `web-app/index.html` supplies the root element, and `web-app/src/main.tsx` mounts the React component tree. `main.tsx` also imports the packaged fonts before rendering `App`. At the root of the component tree, `App.tsx` creates a `BrowserRouter` and delegates application coordination to `RoutedApp`.

`RoutedApp` is the principal state and orchestration component. It owns the selected robot and simulation, the live and database refresh intervals, the emergency-stop state, and the data returned by the application’s custom hooks. It also determines which polling operations and page controls are enabled for the active route. The component combines unit identifiers discovered from real-time files with unit identifiers returned by database simulation options, which permits the same selector to represent both currently reporting and historically recorded units.

The shared application frame is implemented by `CommandShell` in `App.tsx`. It renders the top bar, desktop sidebar, responsive mobile navigation, page-control dock, global motion-inhibit control, and the main content region. React Router maps the principal paths to `DashboardPage`, `MapPage`, `FleetPage`, `LogAnalysisPage`, and `HelpCenterPage`. This routing model retains one common stateful shell while replacing only the page content, which avoids duplicating selectors, navigation, and global controls.

Most page-level and shared presentational components remain in `App.tsx`, including metric cards, panels, live-image rendering, goal-route controls, line charts, loading states, and the Markdown renderer used by the Help Center. Two larger concerns are separated into dedicated modules. `LiveMap.tsx` renders the warehouse as an SVG in compact and expanded variants, while `SimulationTable.tsx` provides reusable simulation, event, and event-telemetry tables together with event-type filtering. Visual rules for the application are centralized in `App.css`, including desktop and mobile layouts, cards, tables, map symbols, and responsive behavior.

The resulting frontend has clear component-level boundaries for mapping and database tables, but it remains deliberately centralized around `App.tsx`. This is consistent with the scale of a prototype and makes the overall control flow directly traceable. It also means that routing, page composition, polling coordination, state ownership, charts, help rendering, and several reusable components share one large module rather than being divided into feature-specific packages.

4.2.3 Client Data and State Management

The frontend uses React state and custom hooks instead of an external state-management library. `useRealtimeUnits`, `useRobotData`, and `useFleetRobotData` implement discovery and polling of live unit data, whereas `useSimulationOptions` and `useSimulationTables` maintain database-backed selections and snapshots. Polling is route-aware: live or database requests are disabled when the current page does not consume the corresponding data. Timeouts are scheduled after each completed request rather than through overlapping fixed intervals, and cleanup functions cancel scheduled work when a component or route changes.

`web-app/src/data.ts` defines refresh interval types, table column metadata, response validators, and fetch functions for real-time units, robot telemetry, simulation options, and database snapshots. The real-time payload contract is represented by `RobotData` in `web-app/src/types.ts`; it includes the unit and simulation identifiers, pose, GPS coordinates, control errors, current and target velocities, current goal, and next planner point. Responses are checked at runtime before they enter React state, so successful HTTP transport alone is not treated as evidence of a valid payload.

Goal-related browser logic is isolated in `web-app/src/goals.ts`. This module validates the complete goal configuration and provides typed helpers for loading, creating, moving, and removing route entries. `useGoalOrder` in `App.tsx` serializes user interaction at the interface level, exposes loading and saving states, and replaces its local state only with the configuration confirmed by the API. Browser `localStorage` is used narrowly for the selected robot and refresh intervals; simulation choice and fetched operational data remain transient React state.

The layered organization of these components is summarized in figure 3. The diagram represents dependencies within the current Web App and the boundary through which it reaches project-owned data and services; the wider system interactions are reserved for the subsequent subsection.

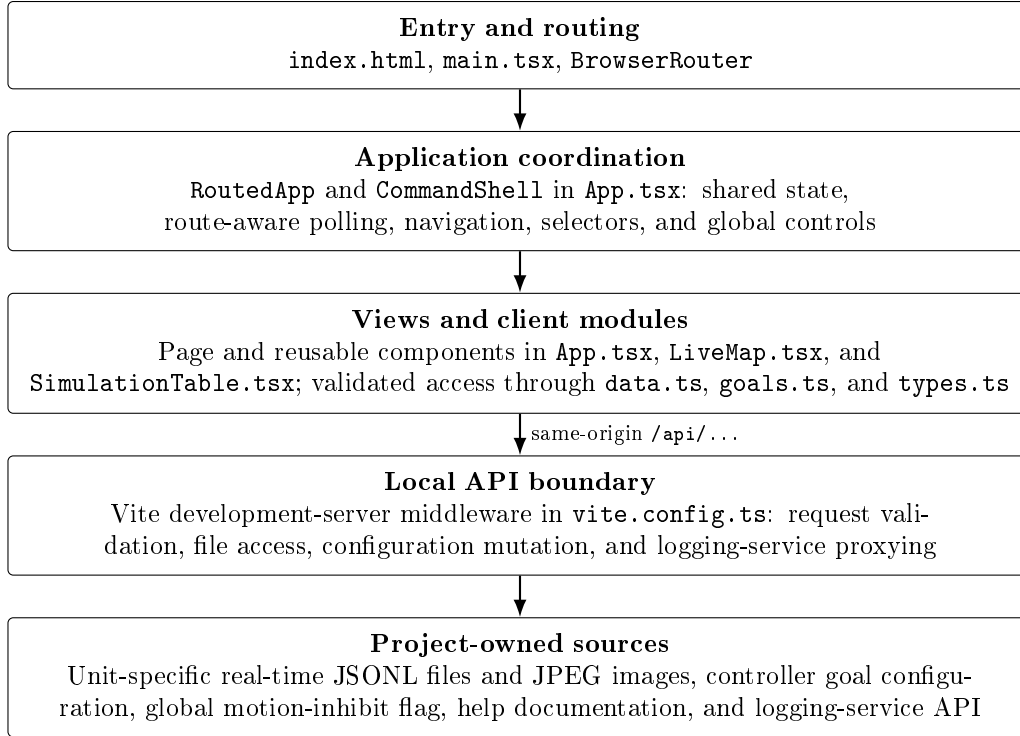


Figure 3: Layered organization of the current React/Vite Web App.

4.2.4 Local API and Persistence Boundary

The server-facing portion of the Web App is implemented as a custom Vite plugin in `web-app/vite.config.ts`. Its middleware exposes endpoints for discovering robots, retrieving a selected unit’s real-time record, serving robot-produced images, loading and mutating goals, reading and changing the application-level global motion inhibit, retrieving the embedded user guide, and proxying database queries. Responses for changing or frequently updated resources are marked with `Cache-Control: no-store`, while method and query validation reject requests outside the expected contract.

File-backed access is unit-aware. The middleware discovers files matching `robot_controller_runs_<id>_re`, normalizes numeric identifiers, and accepts a unit only when the file contains one valid record whose `unit_id` matches the filename. Camera and local-planner requests are likewise resolved to unit-specific JPEG paths. These checks prevent one selected robot from silently receiving another unit’s live payload or image.

Configuration mutations receive stronger server-side checks because they modify controller-owned state. The `/api/goals` handlers validate request bodies and the complete `Goals/RobotRoutes` structure, preserve route invariants, and replace `goals.config.json` through a temporary file and atomic rename. The `/api/emergency-stop` endpoint similarly represents the global inhibit through the presence or absence of `logging/logs/emergency_stop.flag`. The browser there-

fore issues intent through HTTP, while filesystem paths and mutation rules remain confined to the local middleware.

Database-backed requests follow a different path. The `/api/database-snapshot` and `/api/simulation-opt` middleware routes forward their query strings to `logging/logging_server.py`, using `LOGGING_SERVER_URL` when configured and `http://127.0.0.1:8080` otherwise. The Web App consequently does not contain database credentials or issue SQL directly; it consumes the logging service’s JSON interface. This preserves a service boundary between presentation concerns and persistent-data ownership even though the Vite middleware remains local to the frontend development process.

4.2.5 Map Representation and Current Limitation

The map subsystem separates rendering from static geometry. `LiveMap.tsx` receives a validated `RobotData` object and combines the current pose, goal, and next planner point with warehouse bounds, fixed obstacles, and named locations exported by `mapData.ts`. It converts warehouse Y coordinates to the downward-positive SVG coordinate system and derives zoomed view boxes around the selected robot. The component does not perform network requests, which keeps it reusable between the Dashboard and Map View.

The static geometry in `web-app/src/mapData.ts` is, however, an independent duplication of warehouse information also represented elsewhere in the project. No API or generation step synchronizes its bounds, points, or obstacle rectangles with the Webots world or controller configuration. The current display is therefore correct only while these sources remain manually aligned, and a change to the simulated warehouse can leave the Web App map inconsistent without producing a runtime error. This is an architectural limitation of the present prototype rather than an unverified defect in the displayed coordinates.

4.3 Role of the Web App in the Overall Architecture

4.3.1 Architectural Position and Dependencies

The Web App forms the human-facing observation and configuration layer of the AGV system. It does not participate in localization, perception, planning, or low-level control, all of which remain inside each `DefaultController` process. Instead, it translates controller outputs and logging records into operator-oriented views and translates a limited set of user actions into shared configuration or inhibit state. Its position is therefore adjacent to the control loop rather than inside it.

The application depends on three categories of project-owned resources. First, running controllers produce unit-specific real-time telemetry and image artifacts under `logging/logs`. Second, `logging/logging_server.py` provides the persistent read interface backed by MySQL. Third, the controller-owned `goals.config.json` file and the shared motion-inhibit flag support the two control-oriented interactions exposed by the interface. The local Vite middleware in `web-app/vite.config.ts` mediates all three categories, so the browser communicates through same-origin HTTP endpoints rather than through direct filesystem, database, or controller access.

This role makes the Web App a consumer of several subsystems without making it their owner. The controllers own the current robot state and planned route execution, the logging infrastructure owns committed simulation records, and Webots remains the simulation runtime. The Web App owns presentation state such as the current page, selected unit and simulation, and refresh intervals. This separation is visible in the code because no React module imports controller or logging Python code, and no browser request issues SQL or invokes Webots APIs directly.

4.3.2 Live Monitoring Flow

Live telemetry originates in the control cycle of `AGVSimulation.run()` in `DefaultController.py`. After perception, localization, planning, safety limiting, and command calculation, the controller calls `RobotLog.capture_telemetry()` with the resulting state, current goal, command, and next low-level waypoint. `logging/logger/robot_log.py` serializes this information into the unit-specific file `robot_controller_runs_{unit_id}_realtime.jsonl`. The file contains one current record rather than a history and is atomically replaced at intervals of at least 0.05 s of simulation time.

The Vite endpoint `/api/realtime-units` scans these filenames and accepts only nonempty, single-record files whose payload identifier matches the normalized identifier in the filename. `/api/realtime` then resolves one validated unit identifier to its corresponding record. In the browser, `useRealtimeUnits` refreshes unit discovery every three seconds, while `useRobotData` and `useFleetRobotData` poll selected or discovered units at the operator-configured live interval. `data.ts` validates the returned structure against the expected `RobotData` fields before the value is stored and rendered.

The live image paths bypass `RobotLog` but follow the same unit-based identity convention. The perception implementation in `perception/vision.py` writes `camera_feed_{unit_id}.jpg`, while the low-level planner visualization in `planning/low_level/grid_planner.py` writes `local_planner_grid`. The Vite routes `/api/camera-feed` and `/api/local-planner-grid` serve these files with caching disabled, and the React `LiveImage` component includes the selected unit and current time in each request. Telemetry and images can consequently be refreshed independently even though both are produced by the same controller execution.

When a controller stops, its `finally` block calls `RobotLog.clear_realtime()`, leaving the unit's real-time file empty. Subsequent discovery requests exclude that unit unless it remains available through database simulation options, in which case the interface can still expose its committed history but reports its live signal as unavailable. This behavior distinguishes robot presence from historical identity and prevents a stopped robot's final pose from being presented as current.

4.3.3 Persistent Data Flow

Persistent data follows the logging architecture described in sections 3.1, 3.3 and 3.6. During controller initialization, `RobotLog.start()` sends `POST /start` to the logging service. The service creates a `Simulations` row in MySQL and returns its positive `sim_id`; if this exchange does not succeed within the configured retries, `DefaultController` stops the motors and refuses to enter normal operation. The database identity is therefore established before the Web App can observe a moving robot.

During the run, the controller retains discrete events and their associated telemetry in memory and normally sends them to `POST /save` every five seconds. Each accepted save updates the existing `Simulations` summary and inserts the corresponding `Events` and `EventTelemetry` rows in one transaction. Event telemetry is therefore a snapshot associated with a logged event rather than a copy of every 0.05 s real-time update. This distinction explains why the live and database-backed displays can show different temporal resolutions without representing contradictory sources.

The Web App retrieves committed data through two Vite proxy routes. `/api/simulation-options` forwards to the logging service to obtain the unit-to-simulation mapping used by the selectors, and `/api/database-snapshot` forwards the selected `unit_id` and optional `sim_id`. The logging service executes the simulations, events, and event-telemetry queries within one read-only consistent snapshot and returns at most the most recent 200 matching rows from each table. `data.ts` validates the combined response before `SimulationTable.tsx`, the Dashboard, or Log Analysis renders it.

This path deliberately retains MySQL as the authoritative source for committed Web App views. The browser does not connect to MySQL directly. Instead, the logging service owns database credentials, filtering, serialization, transaction consistency, and the read limit, while Vite supplies a same-origin boundary for the frontend. The resulting delay of up to a logging flush interval is accepted for historical and tabular data because the separate real-time channel provides the current operational state.

4.3.4 Goal Configuration Flow

Goal management is a bidirectional interaction between the Web App and the controller-owned task configuration. A Dashboard action is converted by `goals.ts` into a GET, POST, PUT, or DELETE request to `/api/goals`. The Vite handler validates the unit identifier, operation, named goals, finite coordinates, and complete per-unit route structure before replacing `AGV_Webots_World_and_Controllers/c` atomically. React then adopts the server-confirmed configuration rather than assuming that the requested change succeeded.

On the controller side, `Task.refresh()` in `task.py` reloads one validated route snapshot at the beginning of every control cycle and retains the current numeric route position within the new route length. This mechanism makes Web App changes visible during a live simulation without a direct command channel or controller restart. If a configuration is temporarily invalid, the controller reports a warning and retains its last valid snapshot for that cycle, while the Vite endpoint normally prevents the interface itself from writing an invalid structure. The design therefore assigns validation to both sides of the shared file boundary.

The architectural effect of a route change remains limited to task definition. The Web App does not calculate a path, select wheel velocities, or bypass perception and safety processing; it changes the ordered goals that the existing controller pipeline subsequently interprets. This preserves controller ownership of motion while allowing an operator to influence mission-level intent.

4.3.5 Global Motion-Inhibit Flow

The other write-oriented interaction is the control labelled *Emergency Stop*. A browser POST to `/api/emergency-stop` causes Vite to create `logging/logs/emergency_stop.flag`, while DELETE removes it and GET reports its presence. Each active `DefaultController` checks the same path once per control cycle after normal command calculation and the moving-obstacle safety limiter. While the flag exists, the controller replaces the linear, angular, and wheel velocities with zeros before sending the command to the motors.

This is an application-level global motion inhibit rather than a selected-robot command. Its global behavior follows from a single shared flag inspected independently by every `DefaultController`; no central fleet coordinator distributes a stop message. The Web App can confirm the filesystem state returned by its API, but that response is not an independent measurement of physical or simulated motor state. Accordingly, the mechanism is an operational software control within the local prototype and not a separate hardware safety channel.

4.3.6 Integration Rationale and Failure Boundaries

The overall integration separates information by required temporal and consistency properties. Unit-specific files support low-latency, overwrite-in-place telemetry and images, whereas the logging service and MySQL support transactional history, filtering, and cross-table consistency. Goal and inhibit files provide simple bidirectional coordination for processes already sharing the same project filesystem. This combination is economical for a local Webots prototype and avoids introducing a message broker or a dedicated production backend.

The same choices define explicit failure boundaries. A missing or malformed real-time record affects live discovery and monitoring but does not remove committed database records. Loss of

the logging-service read path prevents database-backed views from refreshing, while controller-side buffering can tolerate a transient persistence failure after a simulation has started. Goal management and the global inhibit depend on Vite and the controllers sharing the expected filesystem paths, so these interactions are tightly coupled to same-host execution and the current directory layout.

No fleet-level planning or command arbitration is implemented in the Web App. Fleet Status aggregates independently produced unit snapshots, and the global inhibit is the only application action intentionally shared by all active controllers. The Web App’s overall role is therefore supervisory and observational, with narrowly constrained mission-configuration and motion-inhibit capabilities rather than general remote control of the robots.

4.4 Future Developments

4.4.1 Distributed Architecture for Physical Robots

The principal future development is the replacement of the current same-host, shared-filesystem architecture with a distributed architecture suitable for physical robots. Each robot would continue to execute perception, localization, planning, control, and safety functions locally, but it would no longer depend on files shared with the Web App. Communication would instead be divided between two network services: an extended logging server for authoritative and persistent information, and an MQTT server for real-time publication and notification. The Web App would communicate with these services rather than with robot-owned files.

The robot would contain no persistently stored warehouse map or goal configuration. At every startup it would query the extended logging server for the latest versions, validate them, and retain the accepted values in memory for the duration of its operation. Failure to retrieve or validate either resource would prevent the robot from beginning motion. Once initialization succeeds, the control loop would use its in-memory configuration and would therefore remain independent of ordinary network latency.

4.4.2 Extended Logging and Configuration Server

The existing logging server would retain responsibility for simulation and event persistence while becoming the authoritative source for non-real-time system data. Its HTTP API would serve committed logs, robot goals, and the warehouse map to both the Web App and the robots. MySQL would remain behind this service for structured logging data, so neither robots nor browsers would access the database directly. This extension preserves the existing client-server logging boundary while removing the Web App’s and controllers’ dependence on a common directory layout.

The warehouse map would be stored by the server as one versioned JSON configuration containing the coordinate frame, boundaries, fixed obstacles, and named operational locations required by its consumers. Both the Web App and every robot would retrieve this same definition at startup, eliminating the independent geometry currently compiled into `web-app/src/mapData.ts`. The Web App would treat the map as read-only: modifying warehouse geometry would remain a deliberate server-side administrative operation rather than an operator feature. Because such changes are expected to be rare, an updated map version would take effect when the Web App and robots next start, avoiding replacement of navigation geometry during an active mission.

Goals would also become server-owned, but unlike the warehouse map they would remain editable through the Web App. A goal operation would be sent to the server through HTTP, validated, and committed before the server reported success to the browser. Confirmation from the server would mean that the authoritative configuration had changed; it would not yet prove that a robot had received or applied the change. The subsequent distribution step would use MQTT and an explicit robot acknowledgement.

4.4.3 MQTT Real-Time Communication

MQTT, or Message Queuing Telemetry Transport, is a lightweight publish–subscribe messaging protocol. Its server component is normally called a *broker*: publishers send messages to named topics, and the broker forwards each message to every connected client subscribed to the corresponding topic. Publishers and subscribers therefore do not need direct connections to one another. This decoupling is appropriate for a fleet in which robots and browser clients may connect, disconnect, or subscribe to different unit-specific data streams independently.

In the proposed architecture, each robot would publish its current telemetry and images to unit-specific MQTT topics. The Web App would subscribe only to the robots relevant to the active views and would receive these messages through an MQTT connection supported by the browser, such as MQTT over WebSockets. The specific image encoding, message frequency, and payload transport would remain an implementation decision; the architectural requirement is only that real-time images and telemetry pass through the broker rather than a shared filesystem. MQTT would not become an authoritative historical store, because persistent records would remain under the logging server and MySQL.

Robots would also subscribe to topics carrying configuration-change notifications and application commands. After committing a goal modification, the extended logging server would publish a message identifying the changed resource and its new version. A robot receiving that message would retrieve the authoritative goals from the server, validate them, apply them in memory, and publish an acknowledgement. The interface could consequently distinguish a change that was merely saved by the server from one confirmed as received and applied by a particular robot.

The same path would distribute the control labelled *Emergency Stop*, while preserving its classification as an application-level global motion inhibit. The server would record the desired inhibit state and publish the change through MQTT; subscribed robots would apply it locally and acknowledge the resulting state. This network mechanism would remain distinct from the independent safety-rated emergency-stop hardware required by physical robots. If the broker were unavailable after an authoritative change, the server would need to retain the pending notification and retry publication so that a successful commit could not be silently lost.

4.4.4 Proposed Information Flows

The proposed division of responsibilities is shown in figure 4. HTTP communication carries persistent logs, startup configuration, queries, and validated goal mutations. MQTT carries transient live information, change notifications, acknowledgements, and the software motion-inhibit state. The logging server remains the authority for durable information, whereas the broker provides timely delivery without acquiring ownership of that information.

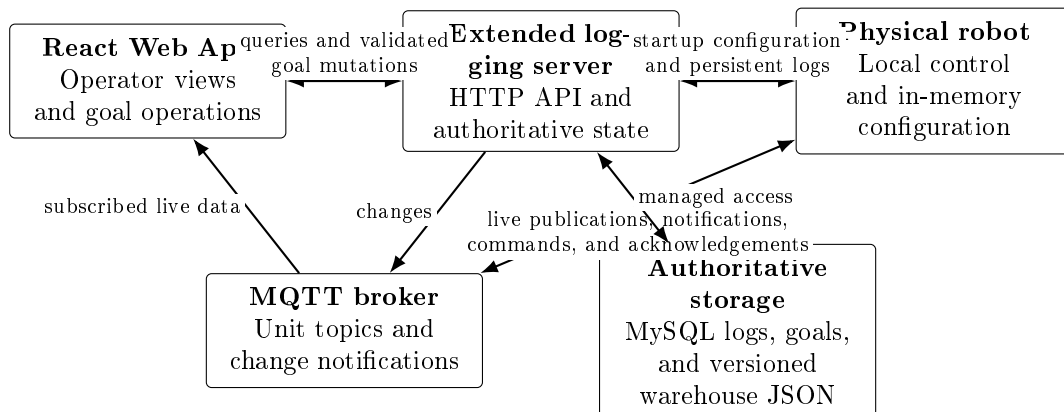


Figure 4: Proposed distributed architecture for the Web App and physical robots.

At startup, the Web App would obtain the current warehouse map, goals, and committed records from the extended logging server before subscribing to the required robot topics. Each robot would likewise obtain and validate the latest map and goals before enabling motion, then publish live telemetry and images through MQTT and persistent event batches through HTTP. This arrangement gives the Web App one authoritative source for non-real-time data and one explicitly transient channel for live monitoring.

The proposed architecture also provides a common integration boundary for simulation and physical deployment. A simulated controller could use the same HTTP and MQTT contracts as a physical robot, allowing the distributed behavior to be tested in Webots before hardware integration. The transition would therefore replace transport and configuration ownership without transferring navigation or safety decisions from the robot to the Web App.